

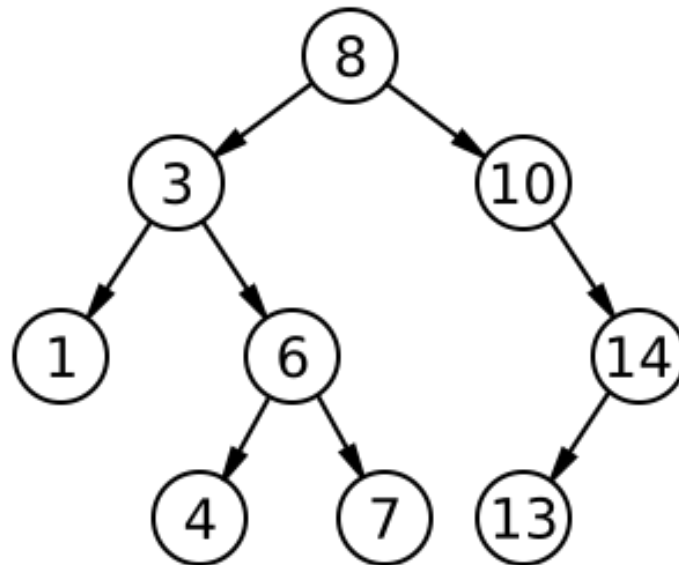
Binary Tree in Data Structure

A structured overview of Binary Tree concepts, types, operations, and application

Introduction to Binary Tree

- A Binary Tree is a hierarchical data structure in which each node has at most two children, referred to as left child and right child.

- Ex:-



Key Terminologies

- Node – Basic element of a tree
- Root – Topmost node
- Edge – Link between nodes
- Leaf – Node with no children
- Subtree – Tree formed by a node and its descendants

Properties of Binary Tree

- Maximum nodes at level 'l' = 2^l
- Maximum nodes in a tree of height 'h' = $2^{(h+1)} - 1$
- Minimum possible height = $\log_2(n+1) - 1$

Types of Binary Trees

- Full Binary Tree – Every node has 0 or 2 children
- Complete Binary Tree – All levels are filled except possibly the last
- Perfect Binary Tree – All internal nodes have two children and all leaves are at same level
- Skewed Binary Tree – All nodes have only one child
- Balanced Binary Tree – Height difference between left and right subtrees is at most one

Representation of Binary Tree

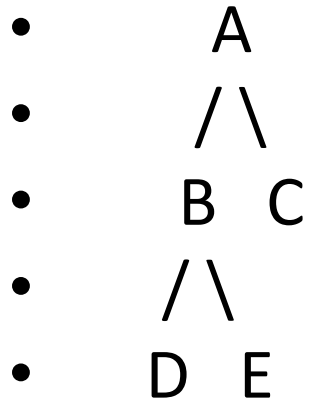
- 1. Linked Representation: Each node contains data, left pointer, right pointer
- 2. Array Representation: Uses index-based relations
 - - For node at index i :
 - Left child = $2i + 1$
 - Right child = $2i + 2$

Tree Traversal Methods

- Inorder (Left, Root, Right)
- Preorder (Root, Left, Right)
- Postorder (Left, Right, Root)
- Level Order (Breadth-First Traversal)

Example of Traversals

- For the tree:



- Traversals:

- • Inorder: D B E A C
- • Preorder: A B D E C
- • Postorder: D E B C A

Binary Search Tree (BST)

- A Binary Search Tree is a binary tree with the property:
- Left child $<$ Root $<$ Right child.
- Operations:
 - Insertion
 - Deletion
 - Searching
 - Traversal

Applications of Binary Tree

- Expression parsing and evaluation
- Hierarchical data representation
- Searching and sorting
- Huffman encoding
- Binary Search Tree for fast lookup

Advantages & Limitations

Advantages:

- Efficient searching and insertion
- Easy to represent hierarchical relationships

Limitations:

- Can become skewed leading to $O(n)$ performance
- Requires balancing for efficiency

Conclusion

- Binary Trees are fundamental in computer science for efficient data organization and retrieval.
- Understanding traversal and BST operations is essential for advanced data structures and algorithms.